

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
Scala For Data Science

Practical XXI

Roll No.: S021	Name: LODHI IQRA
Class: SYBsc IT	Subject: Scala For Data Science
Date of Assignment: 25/08/2026	Date/Time of Submission: 25/08/2026

AIM: Join two CSV files in Spark DataFrames based on a common column and write the output to a file

CODE:

```
ThisBuild / version := "0.1.0-SNAPSHOT"

ThisBuild / scalaVersion := "2.12.18"

libraryDependencies ++= Seq(
  "org.apache.spark" %% "spark-core" % "3.5.1",
  "org.apache.spark" %% "spark-sql" % "3.5.1"
)

lazy val root = (project in file("."))
  .settings(
    name := "IQRA"
  )
```

```

3  object IQRA extends App {
4      val s = SparkSession.builder()
5          .appName("JoinTwoCSV")
6          .master("local[*]")
7          .getOrCreate()
8
9      val d1 = s.read
10         .option("header", "true")
11         .csv("file1.csv")
12
13     val d2 = s.read
14         .option("header", "true")
15         .csv("file2.csv")
16
17     val j = d1.join(d2, Seq("id"), "inner")
18
19     j.show()
20
21     j.write.option("header", "true")
22
23     s.stop()
24     println("IQRA S021")
25 }
26
27

```

Create 2 csv file:

id,name
1,DIVYESH
2,DHARMI
3,SMIT
4,FURVISH
5,HEMISH

id,score
1,85
2,45
3,88
4,78
5,67

OUTPUT:

```
26/08/18 11:41:42 INFO SparkContext: SparkC
+---+-----+-----+
| id|   name|score|
+---+-----+-----+
|  1|DIVYESH|  85|
|  2| DHARMI|  45|
|  3|   SMIT|  88|
|  4| URVISH|  78|
|  5| HEMISH|  67|
+---+-----+-----+

26/08/18 11:41:42 INFO SparkUI: Stopped Spa
26/08/18 11:41:42 INFO MapOutputTrackerMast
26/08/18 11:41:42 INFO MemoryStore: MemoryS
```

Practical XXII

AIM:Set up Apache Spark locally and count the frequency of words in a text file.

CODE:

```

1
2 import org.apache.spark.sql.SparkSession
3
4 object IQRA extends App
5 {
6   val s = SparkSession.builder()
7     .appName("WordCountExample")
8     .master("local[*]")
9     .getOrCreate()
10
11   val r = s.sparkContext.textFile("Input.txt")
12   val w = r.flatMap(line => line.split("\\s+"))
13   val wo = w.map(word => (word, 1))
14   val wc = wo.reduceByKey(_ + _)
15   wc.collect().foreach(println)
16   s.stop()
17   println("IQRA S021")
18 }
19

```

OUTPUT:

```

26/08/25 11:26:37 INFO SparkUI: Stopped S
26/08/25 11:26:37 INFO MapOutputTrackerMa
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(Spark,2)
26/08/25 11:26:37 INFO MemoryStore: Memor
26/08/25 11:26:37 INFO BlockManager: Bloc
26/08/25 11:26:37 INFO BlockManagerMaster
26/08/25 11:26:37 INFO OutputCommitCoordi
26/08/25 11:26:37 INFO SparkContext: Succ
26/08/25 11:26:37 INFO ShutdownHookManage
26/08/25 11:26:37 INFO ShutdownHookManage
IQRA S021

Process finished with exit code 0

```

Practical XXIII

AIM: To filter rows in a CSV file using Apache Spark DataFrame where the numeric Sales column exceeds a specified threshold of 5000.

CODE:

```
1  import org.apache.spark.sql.SparkSession
2
3  object IQRA extends App {
4    val spark = SparkSession.builder()
5      .appName("FilterCSVExample")
6      .master("local[*]")
7      .getOrCreate()
8
9    import spark.implicits._
10
11    val df = spark.read
12      .option("header", "true")
13      .option("inferSchema", "true")
14      .csv("src/main/resources/superstore.csv")
15
16    val f = df.filter($"Sales" > 5000)
17
18    println("Filtered Data:")
19    f.show()
20
21    spark.stop()
22    println("IQRA S021")
23  }
24
25
```

OUTPUT:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID
166	CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BM-11140	Becky Martin	Consumer	United States	San Antonio	Texas	78207	Central	TEC-MA-10000822
510	CA-2015-145352	3/16/2015	3/22/2015	Standard Class	CM-12385	Christopher Martinez	Consumer	United States	Atlanta	Georgia	30318	South	OFF-BI-10003527
684	US-2017-168116	11/4/2017	11/4/2017	Same Day	GT-14635	Grant Thornton	Corporate	United States	Burlington	North Carolina	27217	South	TEC-MA-10004125
2506	CA-2014-143917	7/25/2014	7/27/2014	Second Class	KL-16645	Ken Lonsdale	Consumer	United States	San Francisco	California	94122	West	OFF-SU-10000151
2624	CA-2017-127180	10/22/2017	10/24/2017	First Class	TA-21385	Tom Ashbrook	Home Office	United States	New York City	New York	10024	East	TEC-CO-10004722
2698	CA-2014-145317	3/18/2014	3/23/2014	Standard Class	SM-20320	Sean Miller	Home Office	United States	Jacksonville	Florida	32216	South	TEC-MA-10002412
4099	CA-2014-116904	9/23/2014	9/28/2014	Standard Class	SC-20095	Sanjit Chand	Consumer	United States	Minneapolis	Minnesota	55407	Central	OFF-BI-10001120
4191	CA-2017-166709	11/17/2017	11/22/2017	Standard Class	HL-15840	Hunter Lopez	Consumer	United States	Newark	Delaware	19711	East	TEC-CO-10004722
4278	US-2016-107440	4/16/2016	4/20/2016	Standard Class	BS-11365	Bill Shonel	Corporate	United States	Lakewood	New Jersey	8701	East	TEC-MA-10001047
6341	CA-2017-143112	10/5/2017	10/9/2017	Standard Class	TS-21370	Todd Sumrall	Corporate	United States	New York City	New York	10035	East	TEC-MA-10001047
6426	CA-2016-143714	5/23/2016	5/27/2016	Standard Class	CC-12370	Christopher Conant	Consumer	United States	Philadelphia	Pennsylvania	19120	East	TEC-CO-10004722
6521	CA-2017-138289	1/16/2017	1/18/2017	Second Class	AR-10540	Andy Reiter	Consumer	United States	Jackson	Michigan	49201	Central	OFF-BI-10004995
6627	CA-2014-145541	12/14/2014	12/21/2014	Standard Class	TB-21400	Tom Boeckenhauer	Consumer	United States	New York City	New York	10024	East	TEC-MA-10001127
6827	CA-2016-118689	10/2/2016	10/9/2016	Standard Class	TC-20980	Tamara Chand	Corporate	United States	Lafayette	Indiana	47905	Central	TEC-CO-10004722
7667	US-2016-140158	10/4/2016	10/8/2016	Standard Class	DR-12940	Daniel Raglin	Home Office	United States	Providence	Rhode Island	2908	East	TEC-CO-10001469
8154	CA-2017-140151	3/23/2017	3/25/2017	First Class	RB-19360	Raymond Buch	Consumer	United States	Seattle	Washington	98115	West	TEC-CO-10004722
8489	CA-2016-158841	2/2/2016	2/4/2016	Second Class	SE-20110	Sanjit Engle	Consumer	United States	Arlington	Virginia	22204	South	TEC-MA-10001127
8859	CA-2017-135909	10/13/2017	10/20/2017	Standard Class	JW-15220	Jane Waco	Corporate	United States	Sacramento	California	95823	West	OFF-BI-10003527
9040	CA-2016-117121	12/17/2016	12/21/2016	Standard Class	AB-10105	Adrian Barton	Consumer	United States	Detroit	Michigan	48205	Central	OFF-BI-10000545

26/08/25 11:41:01 INFO SparkContext: SparkContext is stopping with exitCode 0.

Practical XXVIII

AIM:To perform a group-by operation in Spark DataFrames to calculate the average Sales for each Category from a CSV file.

CODE:

```
1
2
3 import org.apache.spark.sql.Session
4 import org.apache.spark.sql.functions._
5
6 ▶ object Demo extends App {
7     val spark = SparkSession.builder()
8       .appName("AverageSales")
9       .master("local[*]")
10      .getOrCreate()
11
12     val df = spark.read
13       .option("header", "true")
14       .option("inferSchema", "true")
15       .csv("src/main/resources/superstore.csv")
16
17     val result = df.groupBy("Category")
18       .agg(avg("Sales").alias("Average Sales"))
19
20     println("Average Sales by Category:")
21     result.show()
22
23     spark.stop()
24     println("IQRA S021")
25 }
26
27
```

OUTPUT:

```
+-----+-----+
|      Category|      Average Sales|
+-----+-----+
|Office Supplies|121.69225531914947|
|      Furniture| 353.4459311957568|
|      Technology| 454.5405475802047|
+-----+-----+

26/08/25 11:48:12 INFO SparkUI: Stopped Spark
```

